Assignment 1- Practical coding assignment using SQL

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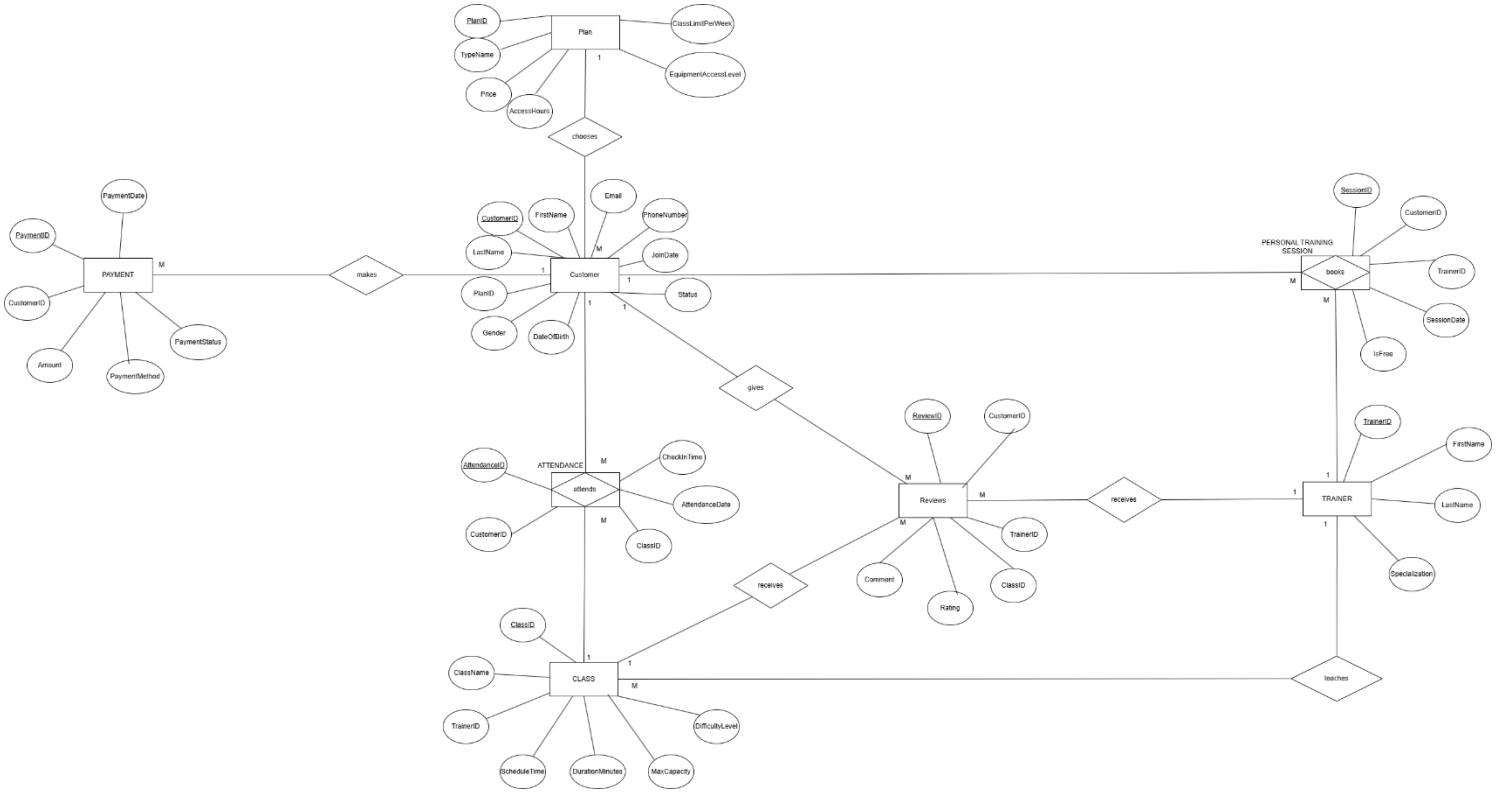
Part A

Database design and implementation

The database was designed for Pulse Fitness, a gym aiming to optimize operations, manage memberships, and support analytical decision-making. It models real-world gym processes including customer subscriptions, payments, class scheduling, personal training sessions, attendance tracking, and feedback collection.

The schema was structured to support both transactional needs (e.g., managing payments and class sign-ups) and analytical insights (e.g., retention, trainer performance, revenue breakdowns).

ERD Diagram



1. Table: Plan

- **Purpose:**

The Plan table stores information about different membership tiers — such as Basic, Premium, and Student. It defines access limits and pricing, which are used to manage customer entitlements and billing.

- **SQL CREATE TABLE Statement:**

```sql
CREATE TABLE Plan (
    PlanID INT PRIMARY KEY,
    TypeName VARCHAR(50),
    Price DECIMAL(6,2),
    ClassLimitPerWeek INT,
    AccessHours VARCHAR(50),
    EquipmentAccessLevel VARCHAR(50)
);
```

- **Sample INSERT INTO Values:**

```sql
INSERT INTO Plan (
    PlanID, TypeName, Price, ClassLimitPerWeek, AccessHours, EquipmentAccessLevel)
VALUES
(1, 'Basic', 30.00, 2, '24/7', 'Limited'),
(2, 'Premium', 50.00, 999, '24/7', 'Full'),
(3, 'Student', 20.00, 3, '9AM-6PM', 'Moderate');
```

- **Explanation:**

This table was implemented first to support foreign key references from Customer. The PlanID is referenced by the Customer table to determine each member's benefits and limits. Fields like Price, ClassLimitPerWeek, and EquipmentAccessLevel are used in analytical queries in Part B (e.g., revenue, usage limits).

2. Table: Customer

- **Purpose:**
  The Customer table stores personal and membership-related information for each gym member. It links each customer to a membership plan and tracks demographic details, contact information, join date, and account status.

- **SQL CREATE TABLE Statement:**

```sql
CREATE TABLE Customer (
    CustomerID INT PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    Gender VARCHAR(10),
    DateOfBirth DATE,
    Email VARCHAR(100),
    PhoneNumber VARCHAR(20),
    JoinDate DATE,
    Status VARCHAR(20),
    PlanID INT,
    FOREIGN KEY (PlanID) REFERENCES Plan(PlanID)
);
```

- **Sample INSERT INTO Values:**

```sql
#Customer
INSERT INTO Customer (CustomerID, FirstName, LastName, Gender, DateOfBirth, Email, PhoneNumber, JoinDate,
VALUES
(1, 'Margaret', 'Sanchez', 'Female', '1987-04-22', 'margaret.sanchez@yahoo.com', '0422355730', '2024-05-24
(2, 'Dana', 'Shaw', 'Male', '1974-04-23', 'dana.shaw@hotmail.com', '0494647234', '2023-11-18', 'Active', 2
(3, 'Thomas', 'Parker', 'Female', '1965-12-13', 'thomas.parker@hotmail.com', '0455444866', '2024-08-29', '
(4, 'James', 'Liu', 'Male', '1984-04-04', 'james.liu@outlook.com', '0410890857', '2023-09-21', 'Active', 3
(5, 'Nicole', 'Tyler', 'Male', '2001-05-02', 'nicole.tyler@gmail.com', '0419823868', '2023-09-24', 'Active
(6, 'Christopher', 'King', 'Female', '1966-08-24', 'christopher.king@yahoo.com', '0470540325', '2024-04-30
(7, 'Matthew', 'Murray', 'Female', '1978-01-08', 'matthew.murray@yahoo.com', '0482350565', '2025-04-27', '
(8, 'Emily', 'Lozano', 'Female', '1982-05-08', 'emily.lozano@outlook.com', '0493395125', '2025-02-18', 'In
(9, 'Mary', 'Perry', 'Female', '1985-10-14', 'mary.perry@hotmail.com', '0486250423', '2024-12-05', 'Active
(10, 'Corey', 'Mitchell', 'Male', '1993-12-12', 'corey.mitchell@gmail.com', '0499386981', '2024-03-23', 'A
```

- **Explanation:**
  The Customer table is central to the schema. It links directly to the Plan table via PlanID to determine access entitlements. It also serves as a foreign key reference for other tables such as Payment, Attendance, PersonalTrainingSession, and Reviews. The inclusion of Status, JoinDate, and Gender supports both operational management and detailed customer analysis in Part B, such as retention rate, engagement, and demographic trends.

3. Table: Trainer

- **Purpose**:
  The Trainer table stores data about personal trainers employed at the gym, including their name and area of specialization. This allows the gym to assign trainers to specific classes and track performance through reviews and personal training sessions.

- **SQL CREATE TABLE Statement:**

```sql
CREATE TABLE Trainer (
    TrainerID INT PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    Specialization VARCHAR(100),
    PhoneNumber VARCHAR(20),
    Email VARCHAR(100)
);
```

- **Sample INSERT INTO Values:**

```sql
#Trainer
INSERT INTO Trainer (TrainerID, FirstName, LastName, Specialization, PhoneNumber, Email)
VALUES
(1, 'Bianca', 'Smith', 'Aqua fitness', '0493464466', 'bianca.smith@trainer.com'),
(2, 'Laura', 'Wilson', 'Cycling', '0443981557', 'laura.wilson@trainer.com'),
(3, 'Robert', 'Hill', 'Yoga', '0424191369', 'robert.hill@trainer.com'),
(4, 'William', 'Cook', 'HIIT', '0479700359', 'william.cook@trainer.com'),
(5, 'James', 'Taylor', 'Core Conditioning', '0424865512', 'james.taylor@trainer.com'),
(6, 'Jasmine', 'Massey', 'Dance', '0471588598', 'jasmine.massey@trainer.com'),
(7, 'John', 'Stephens', 'Body Pump', '0451551900', 'john.stephens@trainer.com'),
(8, 'Matthew', 'Holloway', 'Body Balance', '0477141349', 'matthew.holloway@trainer.com'),
(9, 'Jeffrey', 'Holt', 'Cardio classes', '0421537498', 'jeffrey.holt@trainer.com'),
(10, 'David', 'Wood', 'Stretch & Mobility', '0469159364', 'david.wood@trainer.com'),
(11, 'James', 'Lee', 'Spin class', '0482861151', 'james.lee@trainer.com'),
(12, 'Dominic', 'Torres', 'Indoor cycling', '0411282113', 'dominic.torres@trainer.com'),
(13, 'David', 'Hawkins', 'Strength training', '0440221474', 'david.hawkins@trainer.com'),
(14, 'Krystal', 'Jenkins', 'High-intensity interval training', '0427492930', 'krystal.jenkins@trainer.com
(15, 'Mackenzie', 'Perry', 'Circuit training', '0486235174', 'mackenzie.perry@trainer.com'),
```

- **Explanation:**
  The Trainer table allows the gym to assign instructors to classes (Class.TrainerID) and track who conducts personal training sessions (PersonalTrainingSession.TrainerID) and receives reviews (Reviews.TrainerID). Each trainer specializes in a specific fitness category, which is aligned with the ClassName they teach. This enables analysis on trainer performance, specialization coverage, and customer satisfaction.

4. Table: Class

- **Purpose:**
  The Class table contains information about the group fitness classes offered by the gym. Each class is linked to a specific trainer and includes details such as name, schedule, duration, capacity, and difficulty level. It helps manage gym schedules and analyze participation.

- **SQL CREATE TABLE Statement:**

```sql
CREATE TABLE Class (
    ClassID INT PRIMARY KEY,
    ClassName VARCHAR(100),
    TrainerID INT,
    ScheduleTime TIME,
    DurationMinutes INT,
    MaxCapacity INT,
    DifficultyLevel VARCHAR(20),
    FOREIGN KEY (TrainerID) REFERENCES Trainer(TrainerID)
);
```

- **Sample INSERT INTO Values:**

```sql
#Class
INSERT INTO Class (ClassID, ClassName, TrainerID, ScheduleTime, DurationMinutes, MaxCapacity, DifficultyL
VALUES
(1, 'Aqua fitness', 1, '10:30:00', 45, 21, 'Beginner'),
(2, 'Cycling', 2, '21:30:00', 30, 12, 'Advanced'),
(3, 'Yoga', 3, '12:45:00', 30, 16, 'Beginner'),
(4, 'HIIT', 4, '19:00:00', 60, 13, 'Intermediate'),
(5, 'Core Conditioning', 5, '15:30:00', 30, 30, 'Intermediate'),
(6, 'Dance', 6, '08:45:00', 45, 18, 'Advanced'),
(7, 'Body Pump', 7, '20:45:00', 60, 16, 'Beginner'),
(8, 'Body Balance', 8, '12:15:00', 45, 30, 'Intermediate'),
(9, 'Cardio classes', 9, '18:30:00', 45, 17, 'Beginner'),
(10, 'Stretch & Mobility', 10, '11:15:00', 30, 22, 'Intermediate'),
(11, 'Spin class', 11, '19:30:00', 45, 22, 'Advanced'),
(12, 'Indoor cycling', 12, '19:30:00', 45, 29, 'Intermediate'),
(13, 'Strength training', 13, '14:00:00', 30, 18, 'Advanced'),
(14, 'High-intensity interval training', 14, '13:45:00', 30, 20, 'Intermediate'),
(15, 'Circuit training', 15, '08:00:00', 30, 10, 'Beginner'),
```

- **Explanation**

Each class is assigned to a Trainer and defined by a schedule, difficulty level, and size. The ScheduleTime and MaxCapacity fields are essential for resource planning and managing attendance. This table connects to Attendance (to track who joins), Reviews (for feedback), and supports analysis such as occupancy rates and class popularity in Part B.

5. Table: Attendance

- **Purpose:**
  The Attendance table logs each instance of a customer attending a group fitness class. It records which class was attended, by which customer, on what date, and at what check-in time. This table is key to tracking engagement and analyzing class utilization.

- **SQL CREATE TABLE Statement:**

```sql
CREATE TABLE Attendance (
    AttendanceID INT PRIMARY KEY,
    CustomerID INT,
    ClassID INT,
    AttendanceDate DATE,
    CheckInTime TIME,
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),
    FOREIGN KEY (ClassID) REFERENCES Class(ClassID)
);
```

- **Sample INSERT INTO Values:**

```sql
#Attendance
INSERT INTO Attendance (AttendanceID, CustomerID, ClassID, AttendanceDate, CheckInTime)
VALUES
(1, 1, 10, '2025-02-20', '08:15:00'),
(2, 1, 19, '2025-02-07', '09:00:00'),
(3, 2, 18, '2025-02-01', '12:15:00'),
(4, 2, 8, '2025-02-06', '16:45:00'),
(5, 2, 6, '2025-02-24', '10:45:00'),
(6, 3, 15, '2025-04-08', '17:00:00'),
(7, 3, 9, '2025-04-06', '09:30:00'),
(8, 3, 20, '2025-02-28', '07:00:00'),
(9, 4, 7, '2025-01-30', '13:30:00'),
(10, 5, 8, '2025-04-09', '14:45:00'),
(11, 5, 7, '2025-03-14', '07:15:00'),
(12, 5, 17, '2025-03-12', '06:00:00'),
(13, 6, 7, '2025-02-13', '17:45:00'),
(14, 6, 18, '2025-03-26', '16:45:00'),
```

- **Explanation**:
  This table links Customer and Class, forming a many-to-many relationship. It enables the gym to monitor how frequently members attend, which classes are in demand, and whether members are staying within their weekly class limits (as defined in the Plan table). It's essential for engagement analysis, class popularity ranking, and occupancy rate calculations in Part B.

6. Table: Payment

- Purpose:
  The Payment table records financial transactions made by customers for their membership plans. It includes the date, amount, payment method, and status, enabling revenue tracking and analysis of payment behavior.

- **SQL CREATE TABLE Statement:**

```sql
CREATE TABLE Payment (
    PaymentID INT PRIMARY KEY,
    CustomerID INT,
    PaymentDate DATE,
    Amount DECIMAL(6,2),
    PaymentMethod VARCHAR(20),
    PaymentStatus VARCHAR(20),
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
```

- **Sample INSERT INTO Values:**

```sql
#Payment
INSERT INTO Payment (PaymentID, CustomerID, PaymentDate, Amount, PaymentMethod, PaymentStatus)
VALUES
(1, 1, '2025-03-16', 20.00, 'Bank Transfer', 'Completed'),
(2, 2, '2025-01-20', 50.00, 'Credit Card', 'Completed'),
(3, 2, '2025-03-15', 50.00, 'Cash', 'Completed'),
(4, 3, '2025-04-29', 20.00, 'Cash', 'Completed'),
(5, 4, '2025-03-22', 20.00, 'Credit Card', 'Completed'),
(6, 5, '2025-02-12', 30.00, 'Credit Card', 'Completed'),
(7, 5, '2024-11-18', 30.00, 'Bank Transfer', 'Pending'),
```

- **Explanation:**
  This table connects to Customer through CustomerID and logs every payment transaction. PaymentStatus is especially useful for identifying failed or pending payments, which supports financial oversight and failure analysis. This table was used in Part B to analyze revenue by plan type and detect issues with specific payment methods.

7. Table: PersonalTrainingSession

- Purpose:
  This table records one-on-one personal training sessions between customers and trainers. It tracks whether the session was free or paid, which is especially relevant for Premium plan members who are entitled to one free session per month.

- **SQL CREATE TABLE Statement:**

```sql
CREATE TABLE PersonalTrainingSession (
    SessionID INT PRIMARY KEY,
    CustomerID INT,
    TrainerID INT,
    SessionDate DATE,
    SessionTime TIME,
    DurationMinutes INT,
    Status VARCHAR(20),
    Goal VARCHAR(100),
    IsFree BOOLEAN,
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),
    FOREIGN KEY (TrainerID) REFERENCES Trainer(TrainerID)
);
```

- **Sample INSERT INTO Values:**

```sql
#PersonalTrainingSession
INSERT INTO PersonalTrainingSession (SessionID, CustomerID, TrainerID, SessionDate, SessionTime, Duratio
VALUES
(1, 2, 7, '2025-04-30', '16:30:00', 45, 'Completed', 'Flexibility', TRUE),
(2, 2, 1, '2025-04-21', '06:00:00', 30, 'Completed', 'Endurance', TRUE),
(3, 8, 4, '2025-03-14', '18:15:00', 60, 'Completed', 'Flexibility', TRUE),
(4, 10, 10, '2025-04-05', '12:00:00', 60, 'Scheduled', 'Rehabilitation', TRUE),
(5, 12, 4, '2025-04-14', '14:00:00', 45, 'Scheduled', 'Endurance', TRUE),
(6, 12, 18, '2025-04-19', '18:00:00', 60, 'Scheduled', 'Flexibility', TRUE),
(7, 13, 1, '2025-03-15', '19:00:00', 30, 'Missed', 'Rehabilitation', TRUE),
(8, 15, 16, '2025-03-25', '16:45:00', 30, 'Missed', 'Endurance', TRUE),
(9, 15, 3, '2025-04-11', '15:00:00', 45, 'Missed', 'Rehabilitation', TRUE),
(10, 17, 8, '2025-03-02', '12:45:00', 45, 'Scheduled', 'Strength', TRUE),
(11, 17, 6, '2025-03-06', '19:15:00', 60, 'Missed', 'Weight Loss', TRUE),
(12, 23, 7, '2025-03-28', '20:15:00', 45, 'Missed', 'Rehabilitation', TRUE),
```

- **Explanation**:
  This table links both Customer and Trainer, capturing an important part of the gym's value offering. The IsFree boolean helps monitor usage of included benefits under Premium plans. This table supports analysis on session utilization and detects whether Premium customers are making full use of their entitlements.

8. Table: Reviews

- **Purpose:**
  The Reviews table collects feedback from customers about their experience with specific classes and trainers. It supports evaluation of trainer performance, class satisfaction, and overall member experience.

- **SQL CREATE TABLE Statement:**

```sql
CREATE TABLE Reviews (
    ReviewID INT PRIMARY KEY,
    CustomerID INT,
    TrainerID INT,
    ClassID INT,
    Rating INT,
    Comment TEXT,
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),
    FOREIGN KEY (TrainerID) REFERENCES Trainer(TrainerID),
    FOREIGN KEY (ClassID) REFERENCES Class(ClassID)
);
```

- **Sample INSERT INTO Values:**

```sql
#Reviews
INSERT INTO Reviews (ReviewID, CustomerID, TrainerID, ClassID, Rating, Comment)
VALUES
(1, 55, 1, 1, 5, 'Helpful feedback from trainer.'),
(2, 51, 4, 4, 5, 'Helpful feedback from trainer.'),
(3, 35, 18, 18, 5, 'Excellent pace.'),
(4, 47, 7, 7, 4, 'Great class!'),
(5, 9, 17, 17, 5, 'Nice environment.'),
(6, 37, 6, 6, 5, 'Would take it again.'),
(7, 14, 6, 6, 4, 'Could be more challenging.'),
(8, 53, 1, 1, 5, 'Loved the energy!'),
(9, 33, 2, 2, 3, 'Could be more challenging.'),
(10, 50, 4, 4, 3, 'Excellent pace.'),
```

- **Explanation:**
  The Reviews table links Customer, Trainer, and Class — allowing detailed feedback to be attributed to specific sessions and instructors. It supports qualitative and quantitative analysis, such as average trainer ratings, most praised classes, and satisfaction trends.

Part B

Analysis of insights derived from query results and recommendations

Q1. Which membership plan generates the most revenue?

Purpose:
To determine which membership plan contributes most to Pulse Fitness's income, helping guide marketing, pricing, and resource decisions.

SQL Summary:
This query joins the Payment, Customer, and Plan tables to calculate the total amount paid by members under each plan, filtering for payments with status Completed.

```sql
SELECT
    p.TypeName AS PlanType,
    SUM(pay.Amount) AS TotalRevenue
FROM Payment pay
JOIN Customer c ON pay.CustomerID = c.CustomerID
JOIN Plan p ON c.PlanID = p.PlanID
WHERE pay.PaymentStatus = 'Completed'
GROUP BY p.TypeName
ORDER BY TotalRevenue DESC;
```

Insight from Result:
The Premium plan generated the highest revenue ($1,750), significantly ahead of Basic ($690) and Student ($420). This shows that despite fewer customers, the higher price point of Premium results in greater revenue contribution.

PlanType	TotalRevenue
Premium	1750.00
Basic	690.00
Student	420.00

Actionable Recommendation:
Focus promotional efforts on increasing Premium plan sign-ups, as they yield the highest ROI. Consider enhancing or bundling Basic and Student plans to drive more upgrades.

Q2. What is the average number of classes attended per customer per month?

Purpose:
To measure overall member engagement by analyzing how frequently customers attend fitness classes.

SQL Summary:
The query calculates the average number of class attendances per customer per month by grouping Attendance records by customer and month, then averaging the monthly totals across all customers.

```sql
SELECT
    ROUND(AVG(MonthlyCount), 2) AS AvgClassesPerCustomerPerMonth
FROM (
    SELECT
        CustomerID,
        DATE_FORMAT(AttendanceDate, '%Y-%m') AS Month,
        COUNT(*) AS MonthlyCount
    FROM Attendance
    GROUP BY CustomerID, Month
) AS MonthlyAttendance;
```

Insight from Result:
On average, customers attend only **1.27 classes per month**, which is relatively low. This could indicate underutilization of memberships or a need to better motivate participation.

	AvgClassesPerCustomerPerMonth
▶	1.27

Actionable Recommendation:
Introduce engagement strategies such as:

- Monthly class challenges or attendance rewards
- Notifications/reminders for upcoming classes
- Surveys to understand participation barriers

Q3. Which trainers receive the highest average review rating?

Purpose:
To evaluate trainer performance and member satisfaction by analyzing review scores left by customers for personal training or group sessions.

SQL Summary:
This query joins Reviews with Trainer and calculates the average rating and number of reviews per trainer, then ranks them in descending order of rating.

```sql
SELECT
    t.TrainerID,
    t.FirstName,
    t.LastName,
    ROUND(AVG(r.Rating), 2) AS AvgRating,
    COUNT(r.ReviewID) AS ReviewCount
FROM Reviews r
JOIN Trainer t ON r.TrainerID = t.TrainerID
GROUP BY t.TrainerID, t.FirstName, t.LastName
ORDER BY AvgRating DESC, ReviewCount DESC;
```

Insight from Result:

- **Top performers** include **Bianca Smith**, **Matthew Holloway**, and **Mary Escobar** — each with a perfect **5.00 average rating** from multiple reviews.

- Other trainers like **James Taylor** (4.33) and **Krystal Jenkins** (4.25) also have consistently high ratings.
- A few trainers, such as **Laura Wilson** and **Gary Herrera**, have ratings around **3.00**, indicating room for improvement.

TrainerID	FirstName	LastName	AvgRating	ReviewCount
1	Bianca	Smith	5.00	3
8	Matthew	Holloway	5.00	2
19	Mary	Escobar	5.00	2
3	Robert	Hill	5.00	1
9	Jeffrey	Holt	5.00	1
17	Richard	Wilkerson	5.00	1
18	Michaela	Spencer	5.00	1
5	James	Taylor	4.33	3
4	William	Cook	4.25	4
14	Krystal	Jenkins	4.25	4
6	Jasmine	Massey	4.00	3
13	David	Hawkins	4.00	3

Result 17

Actionable Recommendation:

- Recognize and incentivize top-rated trainers through internal awards or bonuses.
- Provide coaching or feedback to lower-rated trainers, potentially based on member comments.
- Use this data to match new members with higher-rated trainers, enhancing first impressions.

Q4. What is the attendance rate compared to class capacity?

Purpose:
To evaluate how effectively each class is utilizing its available spots, helping optimize scheduling, room assignments, and instructor resources.

SQL Summary:
The query joins Attendance and Class, groups by ClassID, and calculates:

- Total number of attendances
- Class capacity (MaxCapacity)
- Attendance rate = attendances ÷ capacity

```sql
SELECT
    c.ClassID,
    c.ClassName,
    c.MaxCapacity,
    COUNT(a.AttendanceID) AS TotalAttendees,
    ROUND(COUNT(a.AttendanceID) * 1.0 / c.MaxCapacity, 2) AS OccupancyRate
FROM Class c
LEFT JOIN Attendance a ON c.ClassID = a.ClassID
GROUP BY c.ClassID, c.ClassName, c.MaxCapacity
ORDER BY OccupancyRate DESC;
```

Insight from Result:

- **Body Pump** had the highest occupancy rate (**81%**), followed by **HIIT (77%)** and **Circuit Training (70%)**, showing strong demand and good scheduling.
- Most classes fall in the **20–50% range**, suggesting room for better optimization.
- Classes like **Strength Training (11%)**, **Boxing (12%)**, and **Yoga (13%)** had very low occupancy despite available capacity.

ClassID	ClassName	MaxCapacity	TotalAttendees	OccupancyRate
7	Body Pump	16	13	0.81
4	HIIT	13	10	0.77
15	Circuit training	10	7	0.70
19	Cross-training	18	9	0.50
20	Barre	13	6	0.46
2	Cycling	12	5	0.42
10	Stretch & Mobility	22	9	0.41
17	Pilates	24	9	0.38
1	Aqua fitness	21	7	0.33
9	Cardio classes	17	5	0.29
12	Indoor cycling	29	7	0.24
8	Body Balance	30	7	0.23
11	Spin class	22	5	0.23
6	Dance	18	4	0.22
5	Core Conditioning	30	5	0.17
18	Zumba	29	5	0.17

Result 24

Actionable Recommendation:

- Consider offering high-performing classes (Body Pump, HIIT) at more times or in larger spaces.
- Reevaluate underused classes — consider rescheduling, promoting, or replacing low-performing sessions.
- Use this metric monthly to rebalance the schedule dynamically based on usage trends.

Q5. What is the retention rate by membership plan?

Purpose:
To measure how well each membership plan retains its members over time, helping assess customer satisfaction and plan value.

SQL Summary:
The query identifies customers who are still marked as Active and joined more than 6 months ago. It then calculates the percentage of these retained customers out of the total, grouped by PlanType.

```sql
SELECT
    p.TypeName AS PlanType,
    COUNT(CASE
        WHEN c.Status = 'Active' AND c.JoinDate <= DATE_SUB(CURDATE(), INTERVAL 6 MONTH)
        THEN 1 END) AS RetainedCustomers,
    COUNT(*) AS TotalCustomers,
    ROUND(
        COUNT(CASE
            WHEN c.Status = 'Active' AND c.JoinDate <= DATE_SUB(CURDATE(), INTERVAL 6 MONTH)
        THEN 1 END) * 100.0 / COUNT(*), 2
    ) AS RetentionRatePercent
FROM Customer c
JOIN Plan p ON c.PlanID = p.PlanID
GROUP BY p.TypeName;
```

Insight from Result:

- The **Student plan** has the highest retention rate at **68.42%**, followed by **Premium (65.38%)** and **Basic (53.33%)**.
- This suggests that customers on the Student plan may feel they receive strong value relative to cost.
- Basic plan members have the lowest retention, possibly due to limited features or engagement.

PlanType	RetainedCustomers	TotalCustomers	RetentionRatePercent
Basic	8	15	53.33
Premium	17	26	65.38
Student	13	19	68.42

Actionable Recommendation:

- Analyze what makes Student and Premium plans "stickier" (e.g., benefits, pricing, communication) and apply similar strategies to the Basic tier.
- Consider follow-up or re-engagement campaigns specifically for Basic plan users nearing the 6-month mark.

Q6. Which payment methods have the most failed transactions?

Purpose:
To assess the reliability of different payment methods and identify potential risks to revenue flow.

SQL Summary:
This query filters the Payment table to include only failed transactions (PaymentStatus = 'Failed'), then groups them by PaymentMethod to count the number of failures for each.

```sql
SELECT
    PaymentMethod,
    COUNT(*) AS FailedPayments
FROM Payment
WHERE PaymentStatus = 'Failed'
GROUP BY PaymentMethod
ORDER BY FailedPayments DESC;
```

Insight from Result:

- **Credit Card** payments had the highest number of failures (2), followed by **Debit Card** (1).
- No failures were reported for other methods such as Cash or Bank Transfer.
- The overall failure rate appears low in this dataset, but Credit Cards may warrant closer monitoring.

PaymentMethod	FailedPayments
Credit Card	2
Debit Card	1

Actionable Recommendation:

- Review payment failure reasons for Credit Card transactions (e.g., expired cards, authorization errors).
- Consider implementing reminders or fallback payment methods.
- Monitor trends monthly to ensure failure rates remain low.

Q7. What times of day are most classes attended?

Purpose:
To determine the most popular hours for class attendance, enabling more effective scheduling and facility usage.

SQL Summary:
The query extracts the hour from CheckInTime in the Attendance table and groups results by hour to count total attendances for each time slot.

```sql
SELECT
    HOUR(CheckInTime) AS HourOfDay,
    COUNT(*) AS TotalAttendances
FROM Attendance
GROUP BY HourOfDay
ORDER BY TotalAttendances DESC;
```

Insight from Result:

- The most attended time slots are **5 PM (14 attendances)**, **4 PM (11)**, and **2 PM (11)**, indicating strong afternoon and early evening engagement.
- Early morning slots like **6 AM and 7 AM** also show solid usage (~10 attendances), while late evening hours are less utilized.
- **11 AM** had the lowest activity with just 1 recorded check-in.

HourOfDay	TotalAttendances
17	14
16	11
14	11
7	10
13	10
6	10
12	9
8	8
9	8
20	8
10	7
18	6
15	5
19	5
11	1

Actionable Recommendation:

- Consider expanding afternoon and early evening offerings (2–5 PM).
- Maintain or promote early morning options for working members.
- Reallocate underused time slots (e.g., 11 AM) to higher demand periods or offer specialty sessions.

Q8. Which trainers are underutilized compared to the number of classes they teach?

Purpose:
To identify trainers whose classes are attracting relatively low attendance, helping the gym optimize staffing, scheduling, and trainer performance.

SQL Summary:
This query joins Trainer, Class, and Attendance to:

- Count how many classes each trainer teaches
- Total attendances across those classes
- Calculate **average attendance per class** for each trainer

```sql
SELECT
    t.TrainerID,
    CONCAT(t.FirstName, ' ', t.LastName) AS TrainerName,
    COUNT(DISTINCT c.ClassID) AS ClassesTaught,
    COUNT(a.AttendanceID) AS TotalAttendances,
    ROUND(COUNT(a.AttendanceID) * 1.0 / COUNT(DISTINCT c.ClassID), 2) AS AvgAttendancePerClass
FROM Trainer t
JOIN Class c ON t.TrainerID = c.TrainerID
LEFT JOIN Attendance a ON c.ClassID = a.ClassID
GROUP BY t.TrainerID, t.FirstName, t.LastName
ORDER BY AvgAttendancePerClass ASC;
```

Insight from Result:

- Trainers like **Robert Hill** and **David Hawkins** are significantly underutilized, averaging only **2 attendees per class**.
- **Krystal Jenkins** and **Sara Whitney** also show low engagement (~3 per class).
- In contrast, **John Stephens** averaged **13 attendees per class**, indicating strong demand.
- All trainers appear to teach only **1 class** each, so average attendance is especially telling.

TrainerID	TrainerName	ClassesTaught	TotalAttendances	AvgAttendancePerClass
3	Robert Hill	1	2	2.00
13	David Hawkins	1	2	2.00
14	Krystal Jenkins	1	3	3.00
16	Sara Whitney	1	3	3.00
6	Jasmine Massey	1	4	4.00
2	Laura Wilson	1	5	5.00
5	James Taylor	1	5	5.00
9	Jeffrey Holt	1	5	5.00
11	James Lee	1	5	5.00
18	Michaela Spencer	1	5	5.00
20	Gary Herrera	1	6	6.00
1	Bianca Smith	1	7	7.00
8	Matthew Holloway	1	7	7.00
12	Dominic Torres	1	7	7.00
15	Mackenzie Perry	1	7	7.00
10	David Wood	1	9	9.00

Actionable Recommendation:

- Review scheduling, location, and promotion of classes taught by underperforming trainers.
- Consider redistributing class load toward higher-performing trainers or pairing underutilized ones with popular formats or time slots.
- Provide coaching, observation, or targeted feedback to improve engagement for low-attendance trainers.